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Essentials of Data Science
Laboratory - 2304102L - 2026

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1. Practical 1

1.1. Practice Lab Assignment

1.1.1. Calculate Momentum

1.1.2. Conditional Calculation Based on th...

1.1.3. Days between Two Dates

1.1.4. Reverse a Number

1.1.5. Multiplication Table

1.2. Lab Assignment

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5.1. Practice Lab Assignment

5.1.1. Stacked Plot

1.1.1. Calculate Momentum07:34

Write a program that accepts the mass of an object (in kilograms) and its velocity (in meters per second), then calculates and displays the momentum of the object. The momentum p is calculated using the formula:

$$p = m \times v$$

where:
 m is the mass of the object (in kilograms).
 v is the velocity of the object (in meters per second).

Input Format:

- A single floating-point number representing the mass of the object in kilograms.
- A single floating-point number representing the velocity of the object in meters per second.

Output Format:

- The output will display calculated momentum with appropriate units (kgm/s) (rounded up to 2 decimal places).

Sample Test Cases

calculate...

```
1 mass=float(input())
2 vel=float(input())
3 mom=mass*vel
4 print(f"{mom:.2f}kgm/s")
```

Average time0.013 s13.25 ms

Maximum time0.016 s16.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 112 ms

Expected output

5.0

10.0

50.00kgm/s

Actual output

5.0

10.0

50.00kgm/s

Test case 211 ms

Terminal

Test cases

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1.1.2. Conditional Calculation Based on the Number o... 05:16

Write a Python program that accepts an integer n as input. Depending on the number of digits in n .

Constraints:

 $1 \leq n \leq 999$

Input Format:

- The input consists of a single integer n .

Output Format:

- If n is a single-digit number, print its square.
- If n is a two-digit number, print its square root (rounded to two decimal places).
- If n is a three-digit number, print its cube root (rounded to two decimal places).
- Else print "Invalid".

Sample Test Cases

+

condition...

Submit

```
1 import math
2 n=int(input())
3 if n>=1 and n<=9:
4     print(n*n)
5 elif n>=10 and n<=99:
6     print(f"{math.sqrt(n):.2f}")
7 elif n>=100 and n<=999:
8     print(f"{math.cbrt(n):.2f}")
9 else:
10    print("Invalid")
```

Average time

0.008 s

8.29 ms

Maximum time

0.013 s

13.00 ms

4 out of 4 shown test case(s) f

3 out of 3 hidden test case(s) |

Test case 1 10 ms

Debug

Expected output

9

81

Actual output

9

81

Test case 2 4 ms

Test case 3 13 ms

Terminal

Test cases

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1.1.3. Days between Two Dates

06:41

Write a Python program to calculate number of days between two dates.

Input Format:

- The first line contains the first date in the format `YYYY-MM-DD`.
- The second line contains the second date in the format `YYYY-MM-DD`.

Output Format:

- An integer representing the number of days between the given dates.

Note:

- The first date should always be considered earlier than the second date.

Sample Test Cases

+

noOfDay...

Submit

```
1 from datetime import date
2 d1=input()
3 d2=input()
4 y1,m1,day1=map(int,d1.split('-'))
5 y2,m2,day2=map(int,d2.split('-'))
6 date1=date(y1,m1,day1)
7 date2=date(y2,m2,day2)
8 print((date2-date1).days)
```

Average time

0.008 s
8.25 ms

Maximum time

0.011 s
11.00 ms

2 out of 2 shown test case(s)

2 out of 2 hidden test case(s)

Test case 1 11 ms

Debug

Expected output

Actual output

2014-07-02

2014-07-02

2014-11-02

2014-11-02

123

123

Test case 2 7 ms

Terminal

Test cases

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1.1.4. Reverse a Number

03:07

Write a Python program to reverse the digits of the number and print the reversed number.

Input Format:

- The input is a single integer.

Output Format:

- The output prints a single integer, which is the reversed number.

Sample Test Cases

reverseN...

[Submit](#)

```
1 n=int(input())
2 rev=int(str(n)[::-1])
3 print(rev)
```

Average time

0.007 s

7.20 ms

Maximum time

0.012 s

12.00 ms

2 out of 2 shown test case(s) passed**3 out of 3 hidden test case(s) passed****Test case 1** 12 ms [Debug](#)

Expected output

5367

Actual output

5367

7635

7635

Test case 2 9 ms[Terminal](#)[Test cases](#)[< Prev](#) [Reset](#) [Submit](#) [Next >](#)

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1.1.5. Multiplication Table

02:07

Write a Python program that takes an integer as input and prints the multiplication table for that integer from 1 to 10.

Input Format:

- The first line of input contains an integer that represents the number for which the multiplication table is to be printed.

Output Format:

- Print the multiplication table for the given number in the format:

<number> x <multiplier> = <result>

Note:

- The character "x" is used in the output to represent multiplication.
- Refer to the visible test-cases for more clarity.

Sample Test Cases

+

multiplica...

```
1 n=int(input())
2 for i in range(1,11,1):
3     print(f"{n} x {i} = {n*i}")
4
5
```

Average time

0.006 s

6.25 ms

Maximum time

0.011 s

11.00 ms

2 out of 2 shown test case(s)

2 out of 2 hidden test case(s)

Test case 1 5 ms

Debug

Expected output

Actual output

8

8

8 x 1 = 8

8 x 1 = 8

8 x 2 = 16

8 x 2 = 16

8 x 3 = 24

8 x 3 = 24

8 x 4 = 32

8 x 4 = 32

8 x 5 = 40

8 x 5 = 40

Terminal

Test cases

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1.2. Lab Assignment

1.2.1. Pass or Fail

1.2.2. Fibonacci Series

1.2.3. Pattern - 1

1.2.4. Pattern - 2

2. Practical 2

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4. Practical 4

5. Practical 5

5.1. Practice Lab Assignment

5.1.1. Stacked Plot

5.2. Lab Assignment

5.2.1. Titanic Dataset

5.2.2. Histogram of passenger information ...

1.2.1. Pass or Fail

40:51

Write a Python program that accepts the number of courses and the marks of a student in those courses.

The grade is determined based on the aggregate percentage:

- If the aggregate percentage is greater than 75, the grade is Distinction.
- If the aggregate percentage is greater than or equal to 60 but less than 75, the grade is First Division.
- If the aggregate percentage is greater than or equal to 50 but less than 60, the grade is Second Division.
- If the aggregate percentage is greater than or equal to 40 but less than 50, the grade is Third Division.

Input Format:

- The first input will be an integer n , the number of courses.
- The second input will be n integers representing the marks of the student in each of the n courses, separated by a space.

Output Format:

If the student passes all courses:

- Print the aggregate percentage (formatted to two decimal places).
- Print the grade based on the aggregate percentage.

If the student fails any course (marks < 40 in any course), print:

- "Fail".

Note:

- Aggregate Percentage: Average performance across all courses

passorFa...

Submit

```
1 n=int(input())
2 marks= list(map(int, input().split()))
3 for m in marks:
4     if m < 40:
5         print("Fail")
6         break
7 else:
8     percentage = sum(marks) / n
9     print(f"Aggregate Percentage: {percentage:.2f}")
10 if percentage >= 75:
```

Average time

0.010 s

9.90 ms

Maximum time

0.015 s

15.00 ms

5 out of 5 shown test case(s)

5 out of 5 hidden test case(s)

Test case 1

15 ms

Debug

Expected output

4

55 65 78 89

Aggregate Percentage: 71.75

Grade: First Division

Actual output

4

55 65 78 89

Aggregate Percentage: 71.75

Grade: First Division

Test case 2

Terminal

Test cases

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5.1. Practice Lab Assignment

1.2.2. Fibonacci Series

02:28

Write a Python program that uses recursion to print the first n terms of the Fibonacci series.

Input Format:

- A single integer n representing the number of terms to generate.

Output Format:

- A single line containing the first n terms of the Fibonacci sequence, separated by spaces.

Sample Test Cases

fibonacci...

```
1 def fibonacci(n):
2
3     if n==1:
4         return 0
5     elif n==2:
6         return 1
7     else:
8
9         return fibonacci(n-1) + fibonacci(n-2)
10
11
```

Average time

0.019 s

19.50 ms

Maximum time

0.053 s

53.00 ms

2 out of 2 shown test case(s)

2 out of 2 hidden test case(s)

Test case 1 5 ms

Debug

Expected output

5

0 1 1 2 3

Actual output

5

0 1 1 2 3

Test case 2 9 ms

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5. Practical 5

5.1. Practice Lab Assignment

1.2.3. Pattern - 1

01:18

Write a Python program to print a pattern of asterisks in the form of a right-angled triangle.

Input Format:

- The input is an integer, representing the number of rows in the pattern.

Output Format

- The output should display the pattern of asterisks (*), with each row containing an increasing number of asterisks.

Note: Refer to the displayed test cases for the sample pattern.

Sample Test Cases

+

rightangl...

Submit

```
1 n=int(input())
2 for i in range(1,n+1):
3     print("*"*i)
```

Average time

0.005 s

4.83 ms

Maximum time

0.008 s

8.00 ms

2 out of 2 shown test case(s)

4 out of 4 hidden test case(s)

Test case 1 8 ms

Debug

Expected output

5

* , \n

* , * , \n

* , * , * , \n

* , * , * , * , \n

* , * , * , * , * , \n

Actual output

5

* , \n

* , * , \n

* , * , * , \n

* , * , * , * , \n

* , * , * , * , * , \n

Terminal

Test cases

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5. Practical 5

5.1. Practice Lab Assignment

</> 5.1.1. Stacked Plot

1.2.4. Pattern - 2

03:59

Write a Python program to print a right-angled triangle pattern of numbers.

Input Format:

- The input is an integer, representing the number of rows in the pattern.

Output Format:

- The output should display the pattern of numbers separated by space, with each row containing increasing numbers starting from 1 up to the row number.

Note:

- Refer to the displayed test cases for the sample pattern.

Sample Test Cases

+

numberP...

Submit

```
1 n=int(input())
2 for i in range(1,n+1):
3     for j in range(1,i+1):
4         print(j,end=" ")
5     print()
```

Average time

0.009 s

8.75 ms

Maximum time

0.012 s

12.00 ms

2 out of 2 shown test case(s)

2 out of 2 hidden test case(s)

Test case 1 10 ms

Debug

Expected output

5

Actual output

5

1

1

1 2

1 2

1 2 3

1 2 3

1 2 3 4

1 2 3 4

1 2 3 4 5

1 2 3 4 5

Terminal

Test cases

< Prev

Reset

Submit

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